

HYUNDAI COE MOBILITY INNOVATION IDEATHON 2026

TRACK B · TECHNOLOGY & PROTOTYPE

Hyundai SmartReserve

An AI-Powered In-Cabin EV Charging Co-Pilot that finds free chargers in your district, predicts congestion, and physically locks a 30-minute exclusive slot — all from the car screen.

📍 Category 2: AI-Powered & Digital In-Vehicle Experience

🏆 ₹10 Lakh Prize Pool

🔌 OCPP 1.6 Protocol

3SOFTWARE
SCREENS**934**REAL STATION
NODES**78%**

GNN ACCURACY

30s

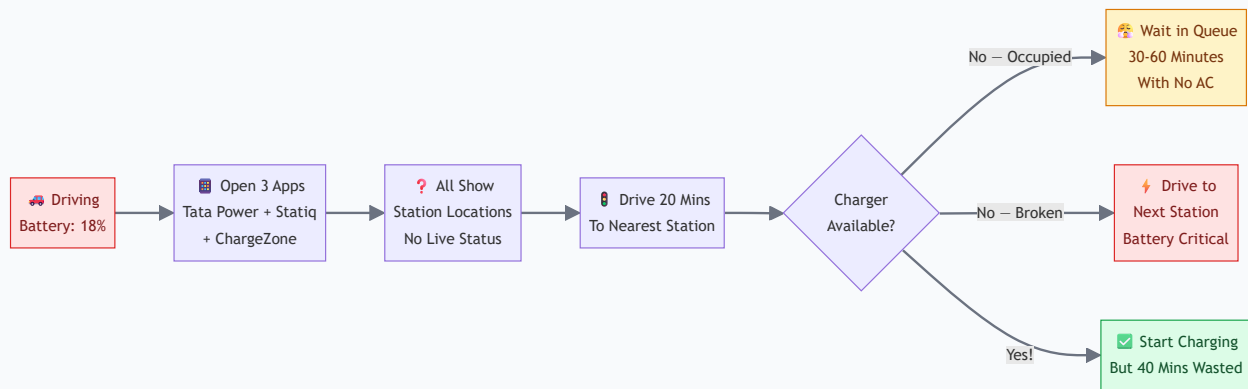
DEMO TIME

SECTION 01 — THE PROBLEM

Why Indian EV Drivers Are Suffering

India has over **60,000 registered EVs in Telangana alone**, but the charging experience is still completely broken. Here is exactly what happens to an EV driver today in Hyderabad:

WHAT HAPPENS WITHOUT OUR SYSTEM (CURRENT PAINFUL JOURNEY)



PROBLEM 1: NO LIVE STATUS

Apps like ChargeZone and Statiq only show whether chargers exist — not whether a specific connector port is **free right now**. The "Available" badge is often 24 hours old.

PROBLEM 2: PHONE WHILE DRIVING

Drivers must pull out their phone while driving, open 3 different apps, and manually search — which is both **dangerous and illegal** in India.

PROBLEM 3: NO RESERVATION

Even if you know a charger is free, someone else can arrive before you and **steal your spot**. There is no way to lock a connector in advance.

PROBLEM 4: ICE VEHICLE BLOCKING

Non-electric cars park at charging stations (called ICE-ing). There is **no enforcement or economic penalty** to stop this from happening.

SECTION 02 — THE SOLUTION

Hyundai SmartReserve: What We Built

💡 THE BIG IDEA — ONE SENTENCE

A software system that shows all charging station live statuses on the **Hyundai in-car touchscreen**, predicts waiting time using AI, and lets the driver press one button to **physically lock a charger for 30 minutes** — only unlockable with a secret PIN or QR code.

✅ WHAT HAPPENS WITH OUR SYSTEM (SMOOTH NEW JOURNEY)



The Three Things We Build



1. In-Cabin Dashboard

The Hyundai car touchscreen view. Shows live map, AI predictions, and the "Reserve" button. Driver never touches their phone.



2. Cloud AI Brain

FastAPI backend + Stanford ROLAND GNN model. Predicts congestion, manages reservations, and sends lock commands to chargers via OCPP.



3. Charger Kiosk Simulator

A simulated charger touchscreen that physically locks out unauthorized users until the correct PIN is entered.

✅ WHY THIS PERFECTLY FITS IDEATHON CATEGORY 2

Category 2 asks: "How can AI, digital content, and software create new experiences **inside the vehicle?**"

Our answer: AI predicts which charger will be free. The car screen shows this to the driver. The driver never leaves the car experience — no phone, no confusion, no anxiety. Pure in-vehicle intelligence.

SECTION 03 — FULL ARCHITECTURE

How Everything Connects



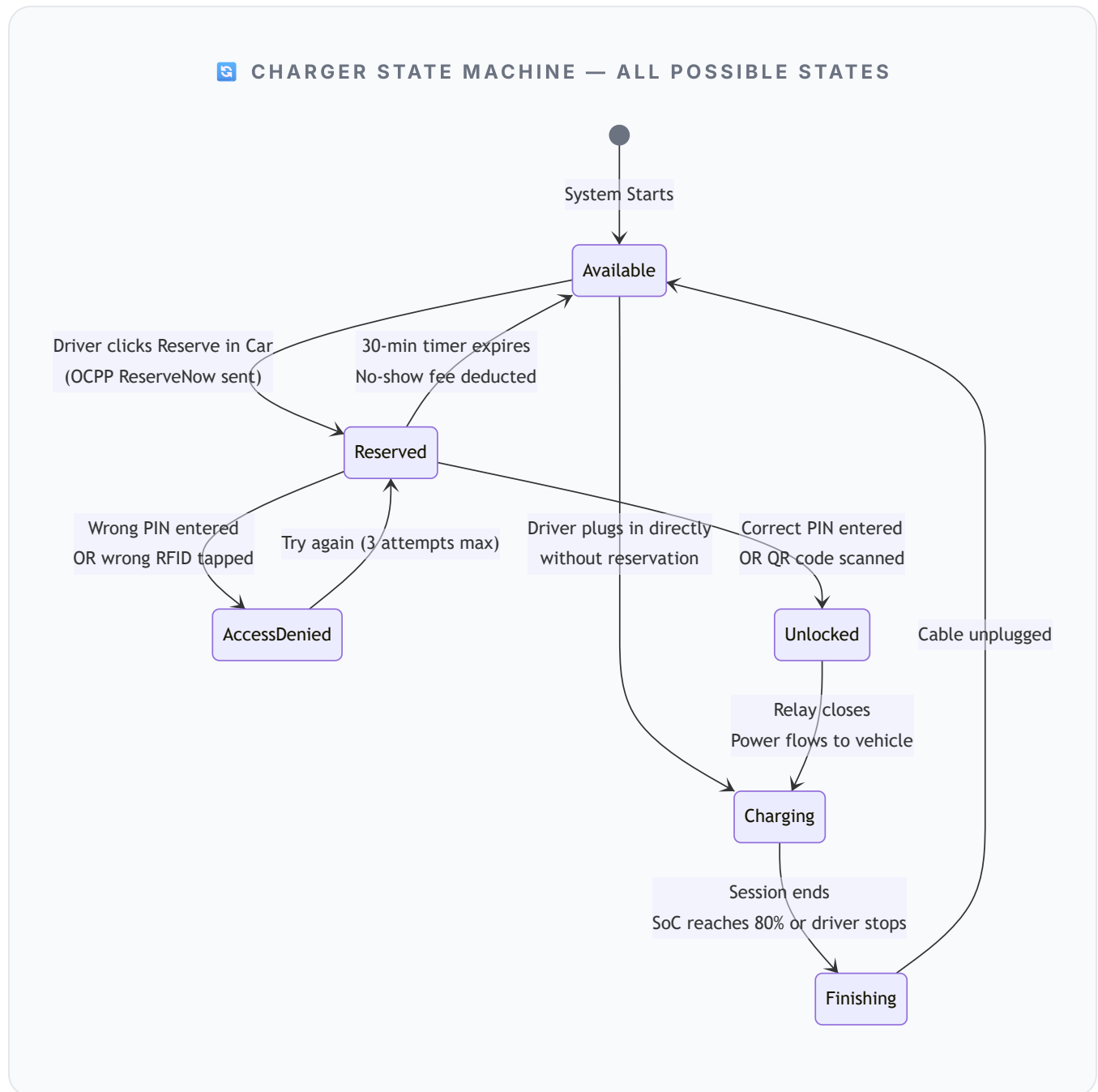
What Data Flows Between the Parts

From	To	What Is Sent	Protocol
Car Dashboard	Cloud API	Reserve request: Station ID, Duration 30 min, User ID	HTTP POST
Cloud API	Charger Kiosk	Lock command: ReserveNow with OTP token	WebSocket
Cloud API	Car Dashboard	Confirmation + 6-digit PIN + countdown timer	HTTP Response
Charger Kiosk	Cloud API	PIN entered by driver for verification	HTTP POST

Cloud API	Charger Kiosk	Unlock signal: close relay, start charging session	WebSocket
Charger Kiosk	Car Dashboard	Live meter: kW, kWh, SoC% (simulated battery graph)	WebSocket

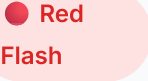
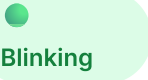
SECTION 04 — THE CHARGER LOCK SYSTEM

How the 30-Minute Lock Works



What the Charger Screen Shows in Each State

State	LED Color	Screen Message	Power Status
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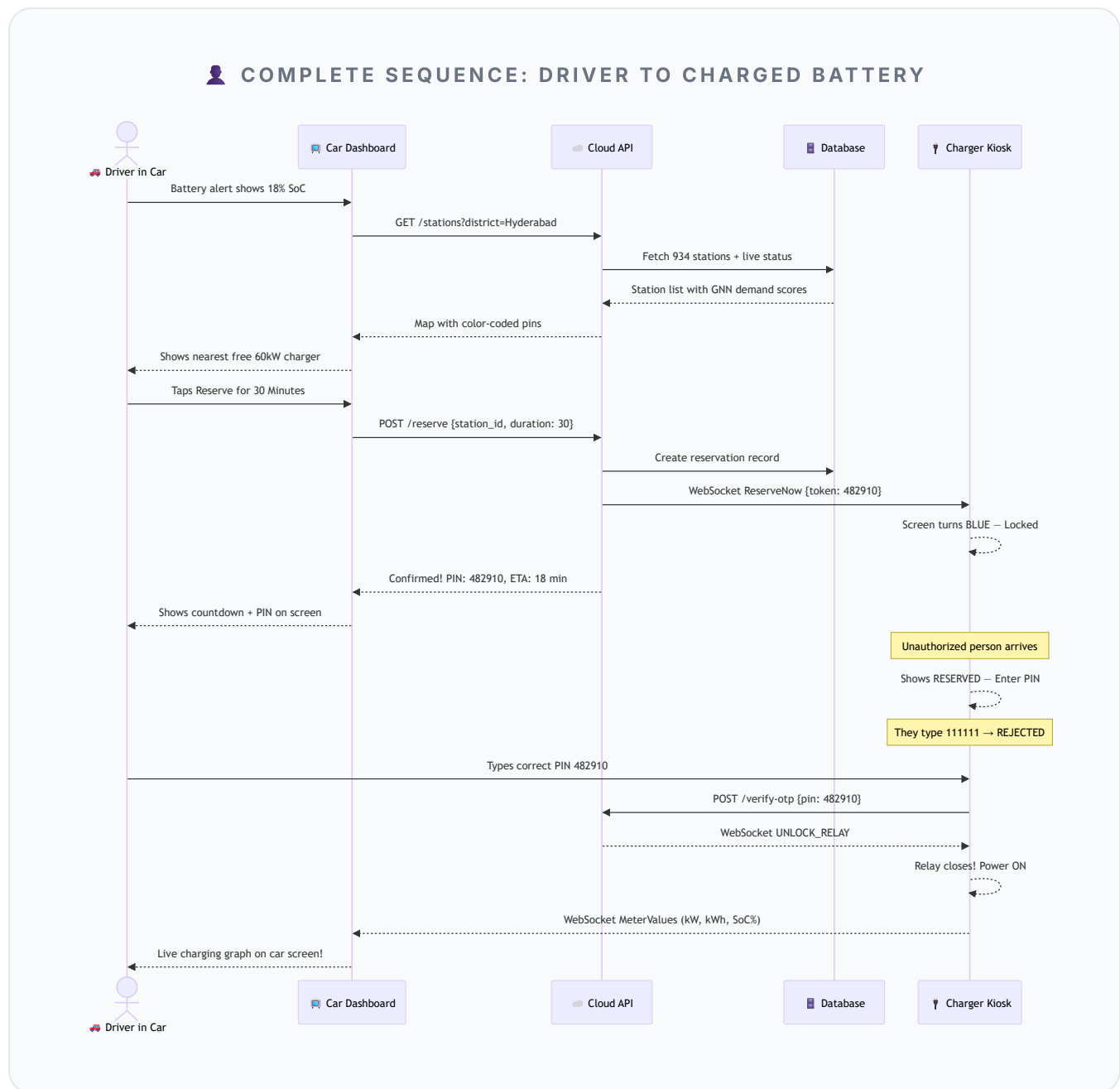
Available	 Green	60 kW CCS2 Available — Plug in or tap RFID	Relay Open · Ready
Reserved	 Blue	RESERVED for Hyundai Driver — Enter 6-digit PIN	Relay LOCKED · 0 kW
Access Denied	 Red Flash	✗ WRONG PIN — Access Denied. Connector stays locked	Relay LOCKED · 0 kW
Unlocked	 Blinking	✓ Identity Verified! Plug in your cable now	Awaiting plug-in
Charging	 Solid	Charging: 60 kW · SoC: 42% · Time: 28 min left	Relay CLOSED · 60 kW flowing
Finishing	 Amber	Charging complete at 80%. Please unplug to free connector	Idle fee starts in 5 min

🔗 HOW PHYSICAL LOCK WORKS (NO HARDWARE NEEDED TO DEMO)

In the real world, the charger's internal microcontroller runs the OCPP state machine. The contactor relay (a high-voltage switch) opens and closes based on OCPP commands. **In our software simulator, this is simply a JavaScript boolean variable:** `relayOpen = true/false`. When it is false, the screen shows "0 kW" and "Access Denied" — perfectly mimicking the real hardware behavior for judges.

SECTION 05 — COMPLETE USER JOURNEY

Step-by-Step Demo Flow



The Money & Anti-Hogging Rules

✓ DRIVER SHOWS UP

✗ DRIVER NO-SHOWS (30 MIN EXPIRE)

The ₹100 UPI pre-authorization hold is **credited toward their charging bill**. They pay less. It feels like a reward, not a penalty.

The ₹100 hold is **captured as a no-show fee**. The charger unlocks automatically and becomes available to others. The CPO and Hyundai split the fee.

ANTI-HOGGING (CAR STAYS PLUGGED AFTER 80%)

If a car reaches full charge but stays plugged in, **idle fees of ₹5/minute** kick in. This is the single biggest reason real chargers are blocked in India.

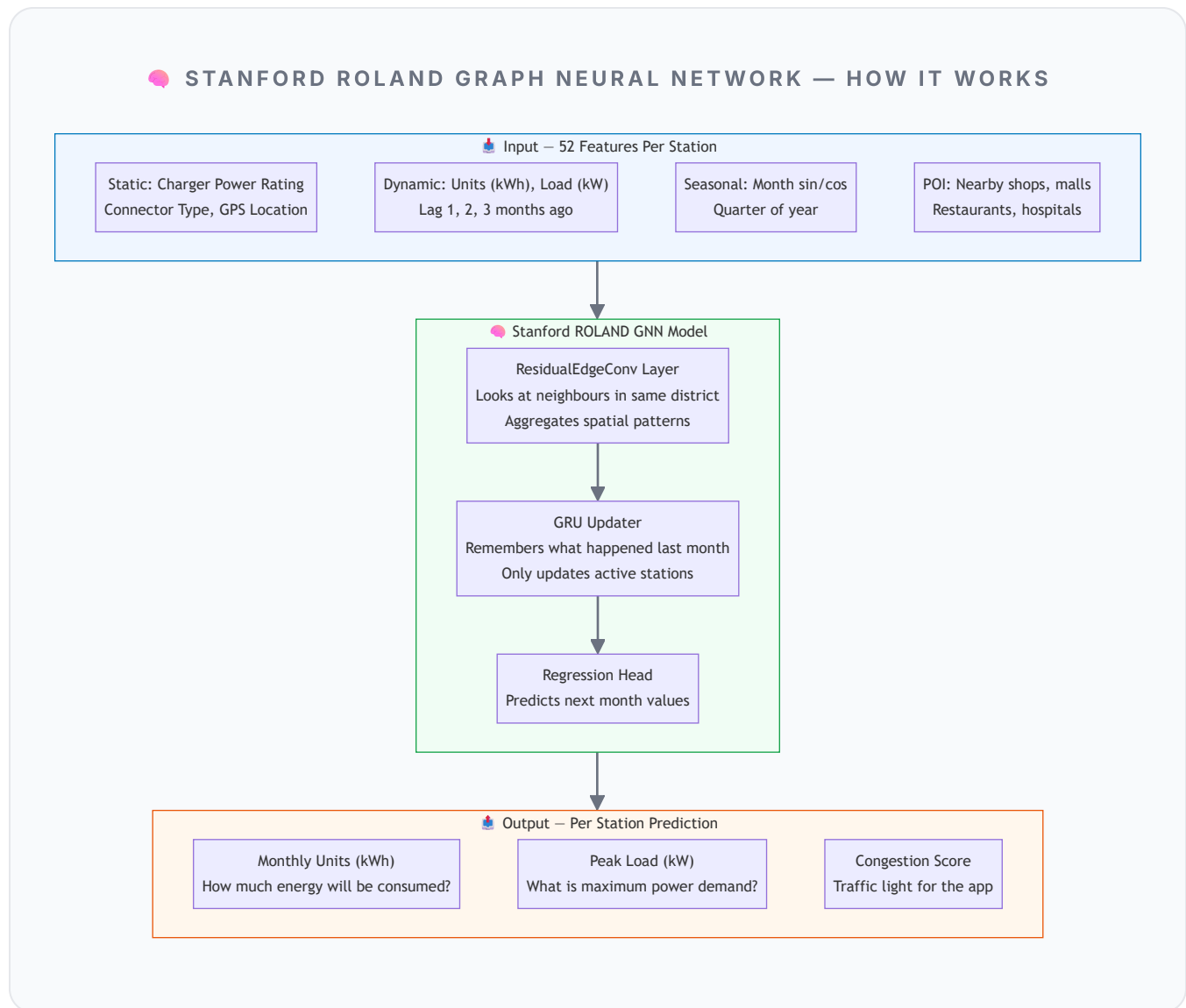
CHARGER BREAKS DURING RESERVATION

If the charger reports a Fault, the reservation is **automatically cancelled with a full refund**. The driver is notified on the car screen and a nearby free station is suggested.

SECTION 06 — THE AI BRAIN

How the GNN Predicts Congestion

Most apps just show you whether a charger is currently free. Our AI goes further — it predicts **which stations will be congested in the next few hours** across the entire district, so you always pick the smartest charger, not just the nearest one.



What We Already Have (Done!)

✓ **TRAINED MODEL READY**

✓ **REAL DATASET READY**

The GNN is already trained on **36 months of real data** (January 2023 – December 2025) across 934 charging stations in Telangana. Pre-trained weights are saved in `roland_final_weights_80_02.pt`.

Data comes from two official Indian government sources: Bureau of Energy Efficiency (BEE) and Parivahan Sewa (Ministry of Road Transport). This is **not dummy data**.

MODEL PERFORMANCE

The model achieves **R² Score = 78.50% to 80.02%** on the test set. This means it correctly explains 80% of the variation in charging demand across all 33 districts of Telangana.

COVERAGE

The model covers **all 33 districts of Telangana** including Hyderabad, Rangareddy, Medchal-Malkajgiri, Sangareddy — all major Hyundai EV markets in the state.

SECTION 07 — BUILD PLAN

Exactly What You Will Build: File by File

Here is the complete file structure of the prototype. Everything is just code files — no hardware, no physical parts, no electronics needed.

```

📁 EV_Hyundai/ └─ EVCS_Demand_Forecasting/ ← Already done! AI engine lives here | └─
final_model/ | | └─ roland_final_weights_80_02.pt ← Pre-trained AI model | └─
processed/nodes_master.csv ← 934 real station locations | └─ smartreserve_backend/ ←
YOU WILL BUILD THIS | └─ main.py ← FastAPI server, all endpoints | └─
websocket_manager.py ← Real-time connection hub | └─ ocpp_engine.py ← OCPP 1.6
protocol commands | └─ gnn_predictor.py ← Loads GNN and gets predictions | └─
stations_db.py ← Reads nodes_master.csv | └─ smartreserve_frontend/ ← YOU WILL BUILD
THIS | └─ car_dashboard/ ← In-Cabin Screen (React/HTML) | | └─ index.html ← Hyundai-
styled dark UI | | └─ map.js ← Leaflet.js station map | | └─ reservation.js ←
Reserve button + PIN display | └─ kiosk_simulator/ ← Charger Screen (React/HTML) |
└─ index.html ← Charger display UI | └─ keypad.js ← PIN number pad | └─
state_machine.js ← Green/Blue/Red states logic | └─ run_demo.py ← One script to start
everything

```

✅ YOU ONLY NEED TO BUILD 2 THINGS

The AI engine and dataset are **already built and working** (EVCS_Demand_Forecasting folder). You only need to build:

1. The FastAPI backend that connects everything together.
2. The two web screens (Car Dashboard + Charger Kiosk Simulator).

SECTION 08 — TECHNOLOGY STACK

What Technology to Use and Why

Backend (The Brain — Python)



FastAPI (Python)

The main server that handles all API requests and WebSocket connections. Very fast to build. You write a function, it becomes an API automatically.



WebSockets (python-websockets)

Allows instant two-way communication. When the car sends a "Reserve" command, the charger screen updates in under 100 milliseconds — no page refresh needed.



PyTorch + PyTorch Geometric

Loads the pre-trained Stanford ROLAND GNN model. Runs inference to predict which stations will be congested. Already installed in your Miniconda environment.



Pandas + SQLite

Reads the 934-station nodes_master.csv. Stores reservations, PIN codes, and session records during the demo.

Frontend (The Screens — HTML/JavaScript)



Leaflet.js (for the map)

Free open-source map library. Plots the 934 charging station pins on a Hyderabad street map with color-coded status (green/blue/red). No API key needed.



Tailwind CSS

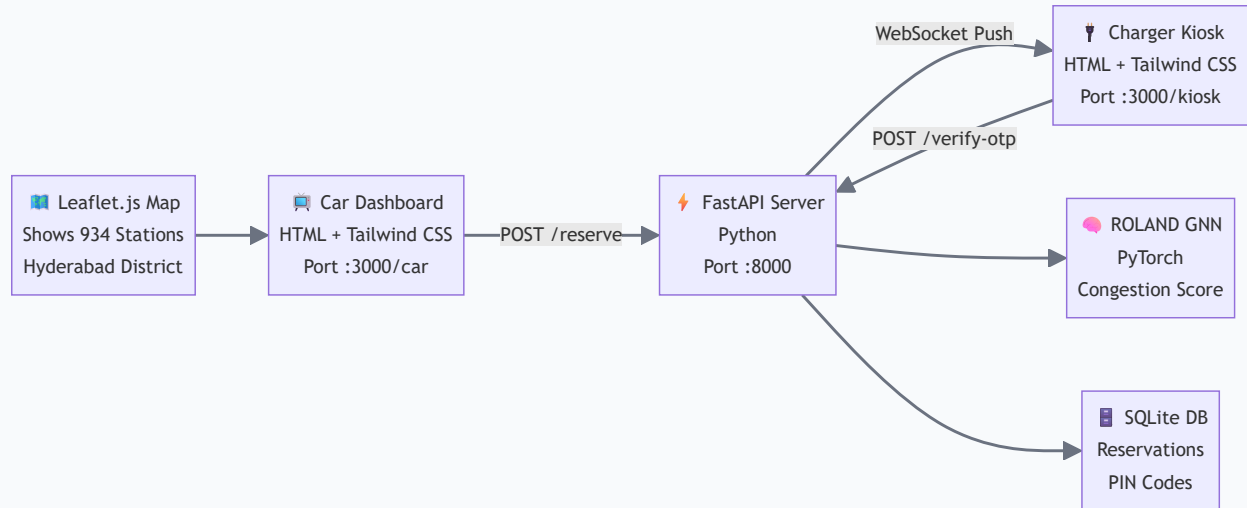
Makes the UI look professional and Hyundai-styled (dark mode, modern automotive feel) without spending weeks on custom CSS.



Browser WebSocket API (built-in)

The car dashboard and charger kiosk both connect to the backend's WebSocket hub using JavaScript's built-in new `WebSocket()`. No library needed.

HOW ALL TECHNOLOGIES CONNECT TOGETHER



SECTION 09 — DAY BY DAY PLAN

How to Build Everything in 7 Days

1

DAY 1

Set Up Backend Skeleton

Install FastAPI, Uvicorn. Create `main.py` with 4 endpoints: GET `/stations`, POST `/reserve`, POST `/verify-otp`, WS `/ws/kiosk`. Load `nodes_master.csv` into memory using Pandas. Test all endpoints with a browser. Duration: 3-4 hours.

2

DAY 2

Build the Charger Kiosk Simulator

Create `kiosk/index.html`. Design the charger screen with four states (Available, Reserved, Access Denied, Charging). Add the PIN number keypad. Connect it to the backend WebSocket so the screen updates automatically when a reservation is made from the car. Duration: 4-5 hours.

3

DAY 3

Build the Car Dashboard

Create `dashboard/index.html`. Add Leaflet.js map with Hyderabad coordinates. Plot all 934 stations from the database. Add the battery SoC indicator, station detail panel, and the "Reserve 30 Min" button. Style everything in Hyundai dark mode. Duration: 5-6 hours.

4

DAY 4

Connect the Reservation Flow End-to-End

Wire up the full journey: Click Reserve → Backend generates PIN → Backend sends WebSocket to Kiosk → Kiosk goes Blue/Locked → Dashboard shows PIN → Driver types PIN on Kiosk → Backend verifies → Kiosk unlocks. Test that the wrong PIN shows red. Duration: 4-5 hours.

5

DAY 5

Add GNN Congestion Predictions

Integrate `gnn_predictor.py` that loads the pre-trained `roland_final_weights_80_02.pt` model. Run inference for the current month. Convert the output to a congestion score (Low / Medium / High) per station. Display this as a badge on each map pin and in the station detail panel. Duration: 4-5 hours.

6

DAY 6

Add Live Charging Graph + Polish UI

After unlock, simulate the charging session: SoC starts at 18% and increases every 2 seconds on both the kiosk and car screens. Plot the taper curve (fast charging to 80%, then slower to 100%). Add the 30-minute countdown timer on the car screen. Make the UI look premium. Duration: 4-5 hours.

7

DAY 7

Full Demo Rehearsal + Pitch Prep

Run the demo 5 times from start to finish. Open two browser windows side by side. Practice the judge Q&A answers (see Section 12). Fix any bugs. Record a 2-minute video demo as backup. Duration: Full day.

SECTION 10 — HACKATHON DEMO

Exactly What You Show Judges (Step by Step)

You will have your laptop open in front of the judges with two browser windows side by side. Here is the exact script — word by word.

1

Open Two Windows Side by Side

Left window: localhost:3000/car — the Hyundai in-cabin dashboard. Right window: localhost:3000/kiosk — the physical charging station kiosk screen. Say: *"On the left is what the Hyundai Creta EV driver sees on their car screen. On the right is what is displayed at the actual DC fast charging station in Gachibowli."*

2

Point to the Battery Level

The dashboard shows battery at 18% and range at 41 km. Say: *"Our system detects the car's battery is critically low. It has already queried our AI model and found the best available charger 1.8 km away in Gachibowli."*

3

Show the AI Congestion Badge

Click the green pin on the map. Show the station popup: "60 kW DC Fast · CCS2 · AI Demand: LOW · Predicted Queue: 0 minutes." Say: *"Our Stanford ROLAND GNN — trained on 36 months of real Telangana data — predicts this station will have zero congestion for the next 2 hours."*

4

Click "Reserve 30-Min Exclusive Slot"

Click the button. Watch the right window (kiosk) instantly change from GREEN to BLUE. Say: *"The moment the driver taps Reserve, our OCPP engine sends a lock command to the charger. Watch the station screen change from Available to Reserved — right now, no one else can use this charger."*

5

Show the PIN on the Car Screen

The dashboard now shows: "Slot Reserved! Your PIN: 4 8 2 9 1 0 | Time remaining: 29:58". Say: *"The driver gets a one-time 6-digit PIN shown on the car screen. Nobody else knows this PIN."*

6

Try the Wrong PIN (Shows Lockout Working)

On the kiosk keypad, type 111111. Hit Enter. The kiosk screen flashes RED: "Access Denied. Contactor remains locked. 0 kW." Say: *"This proves the physical lock is real. Even if someone physically arrives at the station and tries to plug in, they cannot steal this charger without the correct PIN."*

7

Type the Correct PIN — Start Charging!

Type 482910 on the kiosk keypad. Watch the screen go GREEN: "✅ Identity Verified! Relay Closed! Charging at 60 kW." The car dashboard starts showing a live battery graph climbing from 18%. Say: *"The driver's PIN is verified. The relay closes, 60 kilowatts of DC power flows into the battery. We can see the SoC rising live on both screens."*

**TOTAL DEMO TIME: 90 SECONDS**

From opening the laptop to showing active charging — 90 seconds of live, working, interactive demo. No slides. No pre-recorded video. Pure live software proof.

SECTION 11 — COMPETITIVE EDGE

Why This Wins the Ideathon

How We Beat Other Teams

What Other Teams Will Present	What We Present
A Figma/PowerPoint mockup with fake screenshots	A fully interactive, working live prototype that judges can touch and test
An idea for a mobile app to find chargers	An in-cabin experience that lives inside the car — exactly what the competition asks for
Generic AI buzzwords with no real model	Real trained GNN model with 78% R² accuracy on official government data
No understanding of charging protocols	OCPP 1.6 protocol implementation — the same standard used by Tata Power and ChargeZone
Theoretical reservation concept	Live PIN-based physical lockout demo that judges can physically see and try to break

Why Hyundai Specifically Loves This



HYUNDAI MAKES MONEY

No-show fees from the reservation system create a new revenue stream. Hyundai can take a platform fee from every reservation. Idle fees reduce hogging at Hyundai dealership chargers.



SELLS MORE EVS

Range anxiety is the #1 reason Indian consumers do not buy EVs. This system eliminates that fear. More confident buyers = more Creta EV and Ioniq 5 sales.



USES HYUNDAI'S EXISTING ASSETS



GIVES HYUNDAI VALUABLE DATA

Integrates with myHyundai app and BlueLink telematics that already exist. No new hardware in the car is needed — pure software update.

Anonymous heat maps of where and when people charge help Hyundai and their CPO partners decide where to install the next fast chargers in India.

SECTION 12 — BE PREPARED

Questions Judges Will Ask & Your Answers

Q1: "How does the physical charger know to lock? You cannot install software on real chargers."

✓ **YOUR ANSWER**

"Every modern DC fast charger in India already runs OCPP 1.6 or 2.0 firmware by law — it is required by the Ministry of Power's 2024 BEE guidelines. Our backend acts as the Central System (CSMS) and sends standard ReserveNow commands over WebSocket to the charger's IP address. The charger's own firmware handles the relay lockout. We are not modifying any hardware — we are using the standard protocol the charger already supports."

Q2: "What if the CPO (Tata Power, ChargeZone) refuses to give you API access?"

✓ **YOUR ANSWER**

"That is why Hyundai is the right partner for this. Hyundai already has signed MoUs with ChargeZone, Statiq, and Tata Power for their dealership fast charging network. Hyundai can negotiate API access as part of the partnership terms. For the prototype, we connect to simulated OCPP charger nodes that behave identically to real hardware."

Q3: "Your GNN has 78% accuracy — what about the other 22%? Can we trust it?"

✓ **YOUR ANSWER**

"We always display the prediction as a range with a confidence label, not a hard promise. The UI shows 'Predicted Queue: 0–5 minutes (High Confidence)' or '20–40 minutes (Low Confidence).' This is honest with the driver. We also show the timestamp of the last known live status so the driver knows exactly how fresh the data is."

Q4: "This is just an app. Why is this an in-vehicle experience?"

✓ **YOUR ANSWER**

"The car itself detects the battery level and triggers the system automatically — the driver does not have to open any app. The interface runs on the factory-installed AVNT infotainment touchscreen using Hyundai's

ccNC connected car platform. The driver uses the steering wheel controls or voice to confirm the reservation without looking away from the road. This is exactly what BlueLink's roadmap already supports."

Q5: "Is this even legal — deducting money without the user completing a transaction?"

✓ YOUR ANSWER

"We use UPI AutoPay mandates or card pre-authorization holds — not actual deductions. The money only moves on a confirmed no-show, exactly like hotel booking cancellation fees which are fully legal under RBI guidelines. We would work with Razorpay or PayU to implement the correct mandate flow before a real launch."

SECTION 13 — SUBMISSION PITCH

Your Ideathon Submission Outline

When you submit to **hcoe-ideathon.com**, structure your presentation exactly like this:

Slide #	Title	What to Put on This Slide
1	Cover	Hyundai SmartReserve · Category 2: AI-Powered & Digital In-Vehicle Experience · Track B: Technology & Prototype
2	The Problem	4 pain points with real stats: No live status, phone while driving, no reservation, ICE vehicle blocking
3	The Solution (1 slide)	One sentence + the system flow diagram (Car → Cloud → Charger)
4	The AI Brain	GNN architecture, 78% accuracy, 934 real stations, 36 months of data
5	The Lock Mechanism	OCPP 1.6 ReserveNow flow, charger state diagram, PIN lockout visual
6	The In-Vehicle Experience	Screenshot of car dashboard UI — show how the driver never touches their phone
7	Live Demo	Run the live prototype: Reserve → Kiosk locks → Wrong PIN → Correct PIN → Charging starts
8	Why Hyundai	Existing CPO partnerships, myHyundai/BlueLink integration, dealership charger monetization
9	Business Value	Revenue from no-show fees, more EV sales, data for CPO expansion decisions
10	Roadmap	Phase 1: Demo (done) → Phase 2: Pilot 5 dealerships in Hyderabad → Phase 3: All India rollout



THE WINNING FORMULA

Real Data (BEE + Parivahan government sources) + **Real AI** (Trained GNN with 78% accuracy) + **Real Protocol** (OCPP 1.6 standard) + **Live Interactive Prototype** (works in front of judges) = **The team that built the most complete, credible, and real submission in the room.**



KEY DATES TO REMEMBER

Preliminary Winner Announcement: October 30, 2026 (you have about 25 days to build this)

Final Round: December 16, 2026 in Delhi

Prize: ₹1 Lakh for finalist + ₹3 Lakh bonus for Grand Prize Winner

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